Chapter 18

INSTABILITY OF ISOCHRONOUS RINGS

In a storage ring, sometimes there are advantages to work with a lattice having a smaller slippage factor η. One reason is the achievement of shorter bunch lengths. It can be shown easily that, in electron rings where the energy spread is determined by synchrotron radiation, the bunch length is proportional to $|\eta|^{1/2}$. For proton or muon storage rings where there is no synchrotron radiation, the bunch length at fixed rf voltage is proportional to $|\eta|^{1/4}$. Another reason for having a small slip factor is the possible reduction of the expensive rf system. To maintain a bunch at the required rms length σ_τ and momentum spread σ_δ, the synchrotron tune is

$$\nu_s = \frac{|\eta|\sigma_\delta}{\omega_0 \sigma_\tau} \ , \tag{18.1}$$

and the rf voltage is therefore

$$V_{\rm rf} = \frac{2\pi|\eta|\beta^2 E_0 \sigma_\delta^2}{e h \omega_0^2 \sigma_\tau^2 |\cos\phi_s|} \tag{18.2}$$

which decreases linearly as $|\eta|$. In above, h is the rf harmonic, ϕ_s is the synchronous phase angle, E_0 is the total energy of the synchronous particle which has angular revolution frequency ω_0 and velocity βc where c is the velocity of light. Ideally, when $\eta = 0$, no rf will be necessary, because there will not be any drift in phase. A ring with $\eta = 0$, i.e., operating right at transition energy, is called an *isochronous* ring. However, there is always a spread in energy in the beam particles. As a result, it is not possible

for every beam particle to see isochronicity. In addition, the slippage factor η is a nonlinear function of the momentum spread. Usually, isochronicity is defined when the slippage factor vanishes in the first order of the momentum spread. The higher-order contributions will provide a finite slippage. Thus, the ring is actually quasi-isochronous. For such a ring, the parameters of interest are (1) η for the synchronous particle and (2) the total spread in η seen by all the beam particles. It is necessary to design the lattice so that both η and the spread in η are small. When η is vanishing small, there will not be any Landau damping and collective instabilities will emerge as an important issue, which we are going to investigate in this chapter.

18.1 Higher-Order Momentum Compaction

Transition crossing is defined as the moment when the relativistic gamma of the particle is equal to γ_t of the accelerator ring. Let us recall that the transition gamma is defined as $\gamma_t = \alpha_0^{-1/2}$, where α_0 is the momentum-compaction factor which is the fractional increment of the circumferential orbit length of a particle with fractional momentum offset δ. Hence, if $C(\delta)$ is the length of the off momentum orbit,

$$C(\delta) = C_0(1 + \alpha_0\delta) \ , \tag{18.3}$$

with $C_0 = C(0)$ being the length of the on-momentum orbit. Thus, the slippage factor η is exactly zero at transition. However, Eq. (18.3) only gives the linear dependence of the orbit length on momentum offset. In general, this is never the case for any accelerator lattice. Therefore, Eq. (18.3) should be extended to*

$$C(\delta) = C_0\left[1 + \alpha_0\delta(1 + \alpha_1\delta + \alpha_2\delta^2 + \cdots\ )\right] \ , \tag{18.4}$$

where α_1, α_2, etc. are called the high-order components of the momentum-compaction factor. Now the slippage factor η also becomes momentum spread dependent. Its higher orders must be carefully defined so that it enters correctly into Eq. (16.3), the phase-slip equation of motion

$$\frac{d\Delta\phi}{dt} = h\omega_0\eta\delta \ . \tag{18.5}$$

Here, we follow a derivation of Edwards and Syphers [1]. A particle with momentum offset δ_n sees an *accumulated* rf phase ϕ_n on its nth passage of the rf cavity, which is

*In Europe, α_0, α_1, α_2, etc. are usually referred to as α_1, α_2, α_3, etc. There is also another common definition, where $C(\delta) = C_0\left[1 + \alpha_0\delta + \alpha_1\delta^2 + \alpha_2\delta^3 + \cdots\ \right]$.

considered to have an infinitesimal length. On its (n+1)th passage, at a time $T_{n+1}+\Delta T_{n+1}$ later, the accumulated rf phase seen becomes

$$\phi_{n+1} = \phi_n + \omega_{\rm rf}(T_{n+1} + \Delta T_{n+1}) \ , \tag{18.6}$$

where $\omega_{\rm rf}/2\pi$ is the rf frequency, T_{n+1} is the revolution period of the synchronous particle during its (n+1)th turn and ΔT_{n+1} is the extra time taken by the off-momentum particle to complete the revolution. On the other hand, the rf phase seen by the synchronous particle accumulates according to

$$\phi_n^s = \omega_{\rm rf} t_n \ , \tag{18.7}$$

where t_n is the total accumulated time up to the nth passage of the cavity. Naturally, we like to measure the rf phase seen by the off-momentum particle relative to the synchronous particle. This leads to the introduction of the rf phase offset or rf phase slip $\Delta\phi_n$ defined by

$$\Delta\phi_n = \phi_n - \phi_n^s = \phi_n - \omega_{\rm rf} t_n \ . \tag{18.8}$$

Substituting into Eq. (18.6) and noting that $T_{n+1} = t_{n+1} - t_n$, we arrive at

$$\Delta\phi_{n+1} = \Delta\phi_n + \omega_{\rm rf}\Delta T_{n+1} \ . \tag{18.9}$$

In order for the synchronous particle to be synchronized, one must adjust the rf frequency so that $\omega_{\rm rf} T_{n+1} = 2\pi h$ for all turns, where h is the rf harmonic number. Now, we can define the slippage factor as the slip in revolution period at the (n+1)th passage of the cavity by

$$\frac{\Delta T_{n+1}}{T_{n+1}} = \eta_{n+1}\delta_{n+1} \ . \tag{18.10}$$

Here, the subscript of η implies its dependence on the momentum offset of the particle at the (n+1)th passage and *not* its higher-order expansion terms. With this definition, Eq. (18.9) becomes

$$\frac{\Delta\phi_{n+1} - \Delta\phi_n}{T_{n+1}} = \eta_{n+1}\omega_{\rm rf}\delta_{n+1} \ . \tag{18.11}$$

When smoothing is applied, we obtain the phase-slip equation of Eq. (16.10),

$$\frac{d\Delta\phi}{dt} = h\omega_0\eta\delta \ . \tag{18.12}$$

Since the revolution period T can be expressed as

$$T = \frac{C}{\beta c} \ , \tag{18.13}$$

we can easily expand T as a Taylor series in δ, from which each higher-order of the slippage factor can be identified. For example, we have

$$\frac{T_0'}{T_0} = \frac{C_0'}{C_0} - \frac{\beta_0'}{\beta_0} \ ,$$

$$\frac{T_0''}{T_0} = \frac{2{C_0'}^2}{C_0^2} - \frac{2\beta_0' C_0'}{\beta_0 C_0} - \frac{\beta_0''}{\beta_0} + \frac{2{\beta_0'}^2}{\beta_0^2} \ ,$$

$$\frac{T_0'''}{T_0} = \frac{C_0'''}{C_0} - \frac{3\beta_0' C_0''}{\beta_0 C_0} - \frac{3\beta_0'' C_0'}{\beta_0 C_0} + \frac{6{\beta_0'}^2 C_0'}{\beta_0^2 C_0} - \frac{\beta_0'''}{\beta_0} - \frac{\beta_0'\beta_0''}{\beta_0^2} - \frac{6{\beta_0'}^3}{\beta_0^3} \ , \tag{18.14}$$

where the *prime* denotes differentiation with respect to δ and all variables are evaluated at the synchronous particle, which explains why all the variables above carry the subscriptions zero, although these subscripts may have been suppressed in many occasions for the sake of convenience. The derivatives of C can be read off easily from Eq. (18.4). The derivatives of β can be computed straightforwardly. They are:

$$\frac{\beta_0'}{\beta_0} = \frac{1}{\gamma_0^2} \ ,$$

$$\frac{\beta_0''}{\beta_0} = -\frac{3\beta_0^2}{\gamma_0^2} \ ,$$

$$\frac{\beta_0'''}{\beta_0} = -\frac{3\beta_0^2(1-5\beta_0^2)}{\gamma_0^2} \ . \tag{18.15}$$

With the expansion of the slippage factor

$$\eta = \eta_0 + \eta_1\delta + \eta_2\delta^2 + \cdots \ , \tag{18.16}$$

we obtain the expressions for the higher-order components of the slippage factor [Exercise 18.1]:

$$\eta_0 = \alpha_0 - \frac{1}{\gamma_0^2} \ , \tag{18.17}$$

$$\eta_1 = \alpha_0\alpha_1 + \frac{3\beta_0^2}{2\gamma_0^2} - \frac{\eta_0}{\gamma_0^2} \ , \tag{18.18}$$

$$\eta_2 = \alpha_0\alpha_2 + \frac{\alpha_0\alpha_1}{\gamma_0^2} - \frac{2\beta_0^4}{\gamma_0^2} + \frac{3\alpha_0\beta_0^2}{2\gamma_0^2} + \frac{\eta_0}{\gamma_0^4} \ . \tag{18.19}$$

Looking at the phase-slip equation above, one may be tempted to equate $d\Delta\phi/dt$ to $-\Delta\omega/h$, where $\Delta\omega$ is the slip in angular velocity of the off-momentum particle relative to the synchronous particle. However, this will translate the definition of η to

$$\frac{\Delta\omega}{\omega_0} = -\eta\delta \ , \tag{18.20}$$

which is different from Eq. (18.10) and therefore will lead to incorrect expressions for the higher-order terms of η. This misconception comes about in the smoothing procedure from Eq. (18.11) to Eq. (18.12), where we divide throughout by the revolution period of the *synchronous* particle. If $\Delta\omega$ of the off-momentum particle is desired, one should divide instead by $T_{n+1} + \Delta T_{n+1}$, the revolution period of the off-momentum particle. In other words, $d\Delta\phi/dt$ in the phase equation describing the motion of an off-momentum particle does not imply the rate of change of rf phase slip of the off-momentum particle according to the clock that registers the revolution period of that particle. Instead, it is referenced to the clock that registers the revolution period of the synchronous particle. Because of this easily-forgotten detail, it will be more convenient to use $s = v_0 t$ as the independent 'time' variable, where s is the distance measured along the closed orbit of the synchronous particle and v_0 is the velocity of the synchronous particle.

Another definition in the literature is [2]

$$\eta = -\frac{1}{\omega_0}\frac{d\omega}{d\delta} \ , \tag{18.21}$$

which is incompatible with the phase-slip equation in Eq. (18.12). This definition originates from the lowest order expansion in ω [2], and is therefore insufficient when higher-orders in η are studied. This is, in fact, a variation of the incorrect definition of Eq. (18.20).

18.2 η_1-Dominated Bucket

To save the cost of rf power, suggestions have been made to make storage rings *isochronous* or *quasi-isochronous*, implying an operation when $\eta_0 \approx 0$. Since the drift of the longitudinal phase is small, a small rf system will be adequate. However, when η_0 is small enough, we need to include the next lowest nonlinear term of the slippage factor, namely η_1. When the rf phase slip $\Delta\phi$ and the fractional momentum spread δ are used as canonical coordinates with time t being the independent variable, the Hamiltonian describing